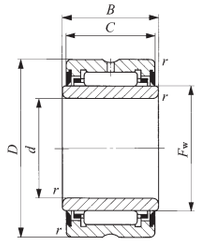
MACHINED TYPE NEEDLE ROLLER BEARINGS

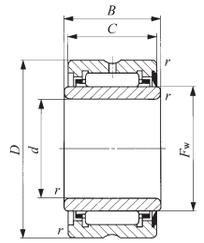
With Seal, With Inner Ring



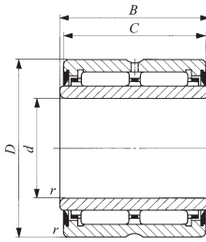




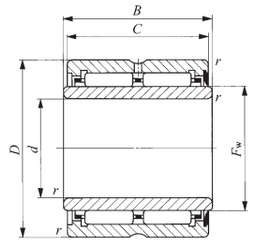
NA49···UU
NA69···UU($d \leqq 30$)



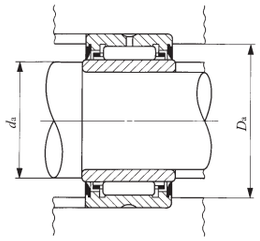
NA49···U
NA69···U($d \leqq 30$)



NA69···UU



NA69···U



Shaft dia. 10 — 40mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: d	D	C	B
10	NA 4900UU	NA 4900U	—	—	24.5	10	22	13	14
12	NA 4901UU	NA 4901U	—	—	27.5	12	24	13	14
	—	—	NA 6901UU	NA 6901U	45.5	12	24	22	23
15	NA 4902UU	NA 4902U	—	—	36	15	28	13	14
	—	—	NA 6902UU	NA 6902U	62.5	15	28	23	24
17	NA 4903UU	NA 4903U	—	—	39.5	17	30	13	14
	—	—	NA 6903UU	NA 6903U	68.5	17	30	23	24
20	NA 4904UU	NA 4904U	—	—	78.5	20	37	17	18
	—	—	NA 6904UU	NA 6904U	137	20	37	30	31
22	NA 49/22UU	NA 49/22U	—	—	87.5	22	39	17	18
	—	—	NA 69/22UU	NA 69/22U	153	22	39	30	31
25	NA 4905UU	NA 4905U	—	—	92.5	25	42	17	18
	—	—	NA 6905UU	NA 6905U	162	25	42	30	31
28	NA 49/28UU	NA 49/28U	—	—	101	28	45	17	18
	—	—	NA 69/28UU	NA 69/28U	177	28	45	30	31
30	NA 4906UU	NA 4906U	—	—	106	30	47	17	18
	—	—	NA 6906UU	NA 6906U	185	30	47	30	31
32	NA 49/32UU	NA 49/32U	—	—	167	32	52	20	21
	—	—	NA 69/32UU	NA 69/32U	300	32	52	36	37
35	NA 4907UU	NA 4907U	—	—	179	35	55	20	21
	—	—	NA 6907UU	NA 6907U	320	35	55	36	37
40	NA 4908UU	NA 4908U	—	—	245	40	62	22	23
	—	—	NA 6908UU	NA 6908U	440	40	62	40	41

$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm: d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm	Assembled inner ring
0.3	14	12	13	20	8 080	8 490	14 000	**LRTZ 101414**
0.3	16	14	15	22	8 470	9 320	12 000	**LRTZ 121614**
0.3	16	14	15	22	15 500	20 400	12 000	**LRTZ 121623**
0.3	20	17	19	26	9 570	11 600	9 500	**LRTZ 152014**
0.3	20	17	19	26	18 500	27 100	9 500	**LRTZ 152024**
0.3	22	19	21	28	10 300	13 100	8 500	**LRTZ 172214**
0.3	22	19	21	28	19 800	30 600	8 500	**LRTZ 172224**
0.3	25	22	24	35	18 000	20 500	7 500	**LRTZ 202518**
0.3	25	22	24	35	33 000	44 600	7 500	**LRTZ 202531**
0.3	28	24	27	37	18 300	23 700	7 000	**LRTZ 222818**
0.3	28	24	27	37	33 800	52 000	7 000	**LRTZ 222831**
0.3	30	27	29	40	20 300	25 100	6 500	**LRTZ 253018**
0.3	30	27	29	40	39 200	58 700	6 500	**LRTZ 253031**
0.3	32	30	31	43	21 000	26 800	6 000	**LRTZ 283218**
0.3	32	30	31	43	38 900	59 100	6 000	**LRTZ 283231**
0.3	35	32	34	45	21 500	28 400	5 500	**LRTZ 303518**
0.3	35	32	34	45	40 100	63 000	5 500	**LRTZ 303531**
0.6	40	36	39	48	29 400	44 200	5 000	**LRTZ 324021**
0.6	40	36	39	48	50 300	88 300	5 000	**LRTZ 324037**
0.6	42	39	41	51	30 100	46 300	4 500	**LRTZ 354221**
0.6	42	39	41	51	51 600	92 600	4 500	**LRTZ 354237**
0.6	48	44	47	58	37 200	58 400	4 000	**LRTZ 404823**
0.6	48	44	47	58	63 700	117 000	4 000	**LRTZ 404841**

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

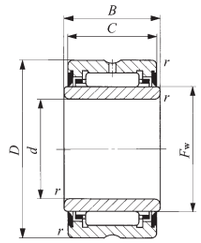
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

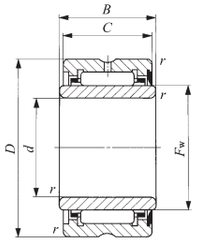
With Seal, With Inner Ring



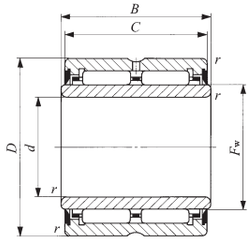




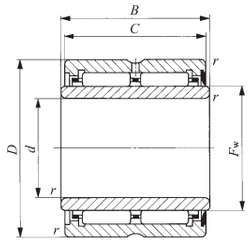
NA49···UU



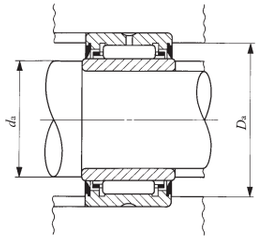
NA49···U



NA69···UU



NA69···U



Shaft dia. 45 — 110mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: d	D	C	B
45	NA 4909UU	NA 4909U	—	—	290	45	68	22	23
	—	—	NA 6909UU	NA 6909U	520	45	68	40	41
50	NA 4910UU	NA 4910U	—	—	295	50	72	22	23
	—	—	NA 6910UU	NA 6910U	530	50	72	40	41
55	NA 4911UU	NA 4911U	—	—	415	55	80	25	26
	—	—	NA 6911UU	NA 6911U	730	55	80	45	46
60	NA 4912UU	NA 4912U	—	—	445	60	85	25	26
	—	—	NA 6912UU	NA 6912U	785	60	85	45	46
65	NA 4913UU	NA 4913U	—	—	475	65	90	25	26
	—	—	NA 6913UU	NA 6913U	845	65	90	45	46
70	NA 4914UU	NA 4914U	—	—	770	70	100	30	31
	—	—	NA 6914UU	NA 6914U	1 400	70	100	54	55
75	NA 4915UU	NA 4915U	—	—	815	75	105	30	31
	—	—	NA 6915UU	NA 6915U	1 480	75	105	54	55
80	NA 4916UU	NA 4916U	—	—	860	80	110	30	31
	—	—	NA 6916UU	NA 6916U	1 570	80	110	54	55
85	NA 4917UU	NA 4917U	—	—	1 300	85	120	35	36
	—	—	NA 6917UU	NA 6917U	2 360	85	120	63	64
90	NA 4918UU	NA 4918U	—	—	1 360	90	125	35	36
	—	—	NA 6918UU	NA 6918U	2 480	90	125	63	64
95	NA 4919UU	NA 4919U	—	—	1 420	95	130	35	36
	—	—	NA 6919UU	NA 6919U	2 600	95	130	63	64
100	NA 4920UU	NA 4920U	—	—	1 980	100	140	40	41
110	NA 4922UU	NA 4922U	—	—	2 150	110	150	40	41

$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm: d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm	Assembled inner ring
0.6	52	49	51	64	38 900	63 400	3 500	**LRTZ 455223**
0.6	52	49	51	64	66 600	127 000	3 500	**LRTZ 455241**
0.6	58	54	57	68	41 300	71 100	3 500	**LRTZ 505823**
0.6	58	54	57	68	70 800	142 000	3 500	**LRTZ 505841**
1	63	60	61	75	52 200	85 700	3 000	**LRTZ 556326**
1	63	60	61	75	89 400	171 000	3 000	**LRTZ 556346**
1	68	65	66	80	54 500	92 800	3 000	**LRTZ 606826**
1	68	65	66	80	93 400	186 000	3 000	**LRTZ 606846**
1	72	70	70.5	85	56 800	99 800	2 500	**LRTZ 657226**
1	72	70	70.5	85	97 400	200 000	2 500	**LRTZ 657246**
1	80	75	78	95	76 900	143 000	2 500	**LRTZ 708031**
1	80	75	78	95	124 000	281 000	2 500	**LRTZ 708055**
1	85	80	83	100	79 600	153 000	2 000	**LRTZ 758531**
1	85	80	83	100	128 000	299 000	2 000	**LRTZ 758555**
1	90	85	88	105	80 700	158 000	2 000	**LRTZ 809031**
1	90	85	88	105	132 000	317 000	2 000	**LRTZ 809055**
1.1	100	91.5	98	113.5	103 000	225 000	1 900	**LRTZ 8510036**
1.1	100	91.5	98	113.5	168 000	448 000	1 900	**LRTZ 8510064**
1.1	105	96.5	103	118.5	106 000	238 000	1 800	**LRTZ 9010536**
1.1	105	96.5	103	118.5	172 000	471 000	1 800	**LRTZ 9010564**
1.1	110	101.5	108	123.5	109 000	250 000	1 700	**LRTZ 9511036**
1.1	110	101.5	108	123.5	177 000	493 000	1 700	**LRTZ 9511064**
1.1	115	106.5	113	133.5	134 000	297 000	1 700	**LRTZ 10011541**
1.1	125	116.5	123	143.5	140 000	322 000	1 500	**LRTZ 11012541**

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication.

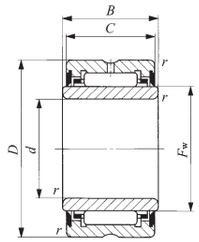
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

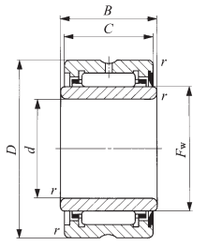
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, With Inner Ring

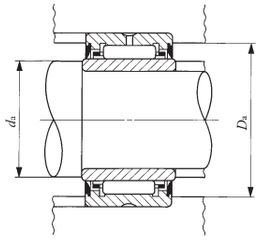




NA49···UU



NA49···U



Shaft dia. 120 — 140mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: d	D	C	B	$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm: d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm	Assembled inner ring
120	**NA 4924UU**	**NA 4924U**	—	—	2 990	120	165	45	46	1.1	135	126.5	133	158.5	178 000	410 000	1 400	**LRTZ 12013546**
130	**NA 4926UU**	**NA 4926U**	—	—	4 080	130	180	50	51	1.5	150	138	148	172	206 000	511 000	1 300	**LRTZ 13015051**
140	**NA 4928UU**	**NA 4928U**	—	—	4 340	140	190	50	51	1.5	160	148	158	182	214 000	549 000	1 200	**LRTZ 14016051**

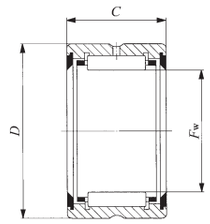
Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

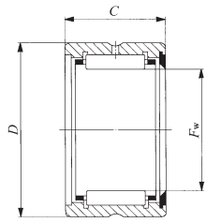
MACHINED TYPE NEEDLE ROLLER BEARINGS

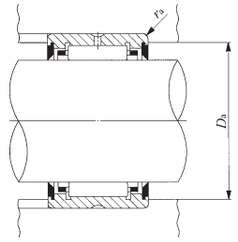
With Seal, Without Inner Ring, Inch Series





BR···UU





BR···U

Shaft dia. 15.875 — 50.800mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			Standard mounting dimensions mm		Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)
	With two seals	With one seal	g	F_w	D	C	D_a Max.	$r_{as\,max}$ (1)	C N	C_0 N	rpm
15.875 (5/8)	BR 101816 UU	BR 101816 U	54	15.875 (5/8)	28.575 (1 1/8)	25.400 (1)	24.5	0.6	18 300	20 000	12 000
19.050 (3/4)	BR 122016 UU	BR 122016 U	68	19.050 (3/4)	31.750 (1 1/4)	25.400 (1)	26.5	1.0	20 700	24 400	10 000
22.225 (7/8)	BR 142216 UU	BR 142216 U	76	22.225 (7/8)	34.925 (1 3/8)	25.400 (1)	29.7	1.0	21 600	26 900	9 000
25.400 (1)	BR 162416 UU	BR 162416 U	83	25.400 (1)	38.100 (1 1/2)	25.400 (1)	32.9	1.0	23 600	31 300	8 000
28.575 (1 1/8)	BR 182620 UU	BR 182620 U	115	28.575 (1 1/8)	41.275 (1 5/8)	31.750 (1 1/4)	36.0	1.0	34 900	49 900	7 000
31.750 (1 1/4)	BR 202820 UU	BR 202820 U	124	31.750 (1 1/4)	44.450 (1 3/4)	31.750 (1 1/4)	39.2	1.0	36 000	53 500	6 500
34.925 (1 3/8)	BR 223020 UU	BR 223020 U	134	34.925 (1 3/8)	47.625 (1 7/8)	31.750 (1 1/4)	42.4	1.0	38 500	60 000	5 500
38.100 (1 1/2)	BR 243320 UU	BR 243320 U	168	38.100 (1 1/2)	52.388 (2 1/16)	31.750 (1 1/4)	45.1	1.5	43 700	66 900	5 500
41.275 (1 5/8)	BR 263520 UU	BR 263520 U	179	41.275 (1 5/8)	55.562 (2 3/16)	31.750 (1 1/4)	48.3	1.5	44 800	70 900	4 500
44.450 (1 3/4)	BR 283720 UU	BR 283720 U	193	44.450 (1 3/4)	58.738 (2 5/16)	31.750 (1 1/4)	51.5	1.5	47 500	78 200	4 500
47.625 (1 7/8)	BR 303920 UU	BR 303920 U	202	47.625 (1 7/8)	61.912 (2 7/16)	31.750 (1 1/4)	54.7	1.5	48 500	82 100	4 000
50.800 (2)	BR 324120 UU	BR 324120 U	216	50.800 (2)	65.088 (2 9/16)	31.750 (1 1/4)	57.8	1.5	51 000	89 400	4 000

Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to grease lubrication.

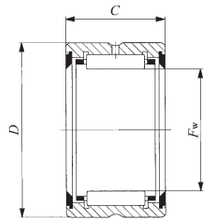
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

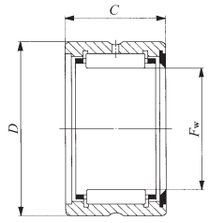
MACHINED TYPE NEEDLE ROLLER BEARINGS

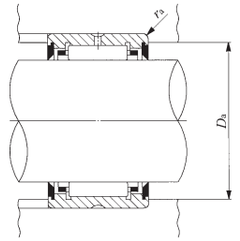
With Seal, Without Inner Ring, Inch Series





BR···UU





BR···U

Shaft dia. 57.150 — 95.250mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			Standard mounting dimensions mm		Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)
	With two seals	With one seal	g	F_w	D	C	D_a Max.	$r_{as\,max}$ (1)	C N	C_0 N	rpm
57.150 (2 1/4)	**BR 364828 UU**	**BR 364828 U**	459	57.150 (2 1/4)	76.200 (3)	44.450 (1 3/4)	69.0	1.5	90 300	158 000	3 500
63.500 (2 1/2)	**BR 405228 UU**	**BR 405228 U**	499	63.500 (2 1/2)	82.550 (3 1/4)	44.450 (1 3/4)	74.3	2.0	94 600	174 000	3 000
69.850 (2 3/4)	**BR 445628 UU**	**BR 445628 U**	540	69.850 (2 3/4)	88.900 (3 1/2)	44.450 (1 3/4)	80.7	2.0	98 700	189 000	2 500
76.200 (3)	**BR 486028 UU**	**BR 486028 U**	585	76.200 (3)	95.250 (3 3/4)	44.450 (1 3/4)	87.0	2.0	105 000	211 000	2 500
82.550 (3 1/4)	**BR 526828 UU**	**BR 526828 U**	891	82.550 (3 1/4)	107.950 (4 1/4)	44.450 (1 3/4)	99.7	2.0	109 000	227 000	2 500
88.900 (3 1/2)	**BR 567232 UU**	**BR 567232 U**	1 098	88.900 (3 1/2)	114.300 (4 1/2)	50.800 (2)	106.1	2.0	142 000	265 000	2 000
95.250 (3 3/4)	**BR 607632 UU**	**BR 607632 U**	1 161	95.250 (3 3/4)	120.650 (4 3/4)	50.800 (2)	111.4	2.5	148 000	287 000	2 000

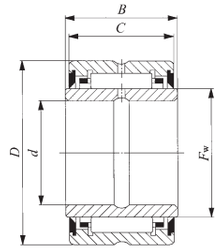
Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

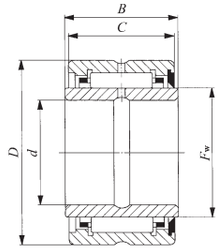
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, With Inner Ring, Inch Series

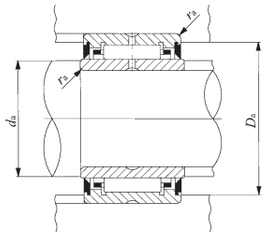




BRI···UU



BRI···U



Shaft dia. 9.525 — 44.450mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			
	With two seals	With one seal	g	d	D	C	B
9.525 (3/8)	BRI 61816 UU	BRI 61816 U	79	9.525 (3/8)	28.575 (1 1/8)	25.400 (1)	25.650
12.700 (1/2)	BRI 82016 UU	BRI 82016 U	99	12.700 (1/2)	31.750 (1 1/4)	25.400 (1)	25.650
15.875 (5/8)	BRI 102216 UU	BRI 102216 U	113.5	15.875 (5/8)	34.925 (1 3/8)	25.400 (1)	25.650
19.050 (3/4)	BRI 122416 UU	BRI 122416 U	127	19.050 (3/4)	38.100 (1 1/2)	25.400 (1)	25.650
22.225 (7/8)	BRI 142620 UU	BRI 142620 U	177	22.225 (7/8)	41.275 (1 5/8)	31.750 (1 1/4)	32.000
25.400 (1)	BRI 162820 UU	BRI 162820 U	196	25.400 (1)	44.450 (1 3/4)	31.750 (1 1/4)	32.000
28.575 (1 1/8)	BRI 183020 UU	BRI 183020 U	211	28.575 (1 1/8)	47.625 (1 7/8)	31.750 (1 1/4)	32.000
31.750 (1 1/4)	BRI 203320 UU	BRI 203320 U	254	31.750 (1 1/4)	52.388 (2 1/16)	31.750 (1 1/4)	32.000
34.925 (1 3/8)	BRI 223520 UU	BRI 223520 U	275	34.925 (1 3/8)	55.562 (2 3/16)	31.750 (1 1/4)	32.000
38.100 (1 1/2)	BRI 243720 UU BRI 243920 UU	BRI 243720 U BRI 243920 U	293 362	38.100 (1 1/2) 38.100 (1 1/2)	58.738 (2 5/16) 61.912 (2 7/16)	31.750 (1 1/4) 31.750 (1 1/4)	32.000 32.000
41.275 (1 5/8)	BRI 264120 UU	BRI 264120 U	386	41.275 (1 5/8)	65.088 (2 9/16)	31.750 (1 1/4)	32.000
44.450 (1 3/4)	BRI 284828 UU	BRI 284828 U	804	44.450 (1 3/4)	76.200 (3)	44.450 (1 3/4)	44.700

F_w	Standard mounting dimensions mm				Basic dynamic load rating C	Basic static load rating C_0	Allowable rotational speed(2)	Assembled inner ring
	d_a Min.	d_a Max.	D_a Max.	$r_{as\ max}$ (1)	N	N	rpm	
15.875 (5/8)	14	14.5	24.5	0.6	18 300	20 000	12 000	LRBZ 61016 B
19.050 (3/4)	17.5	18	26.5	0.6	20 700	24 400	10 000	LRBZ 81216 B
22.225 (7/8)	21	21.2	29.7	0.6	21 600	26 900	9 000	LRBZ 101416 B
25.400 (1)	24	24.4	32.9	0.6	23 600	31 300	8 000	LRBZ 121616 B
28.575 (1 1/8)	27	27.5	36.0	0.6	34 900	49 900	7 000	LRBZ 141820 B
31.750 (1 1/4)	30.5	30.7	39.2	0.6	36 000	53 500	6 500	LRBZ 162020 B
34.925 (1 3/8)	33.5	33.9	42.4	0.6	38 500	60 000	5 500	LRBZ 182220 B
38.100 (1 1/2)	37	37.1	45.1	0.6	43 700	66 900	5 500	LRBZ 202420 B
41.275 (1 5/8)	40.2	40.2	48.3	0.6	44 800	70 900	4 500	LRBZ 222620 B
44.450 (1 3/4) 47.625 (1 7/8)	43.3 43.3	43.4 45	51.5 54.7	0.6 1	47 500 48 500	78 200 82 100	4 500 4 000	LRBZ 242820 B LRBZ 243020 B
50.800 (2)	48	49	57.8	1	51 000	89 400	4 000	LRBZ 263220 B
57.150 (2 1/4)	52.5	55	69.0	1.5	90 300	158 000	3 500	LRBZ 283628 B

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.

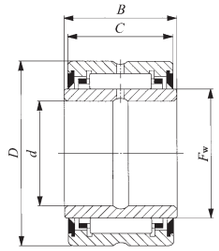
Remarks1. The inner ring and the outer ring each have an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

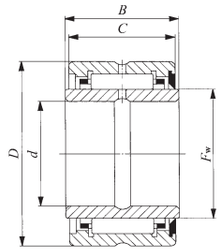
MACHINED TYPE NEEDLE ROLLER BEARINGS

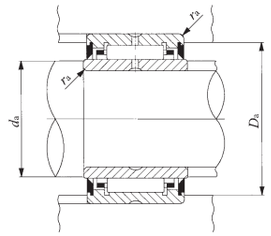
With Seal, With Inner Ring, Inch Series





BRI···UU





BRI···U

Shaft dia. 50.800 — 82.550mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)				F_w	Standard mounting dimensions mm				Basic dynamic load rating C	Basic static load rating C_0	Allowable rotational speed(2)	Assembled inner ring
	With two seals	With one seal	g	d	D	C	B		d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$(1)	N	N	rpm	
50.800 (2)	**BRI 325228 UU**	**BRI 325228 U**	889	50.800 (2)	82.550 (3 1/4)	44.450 (1 3/4)	44.700	63.500 (2 1/2)	58	61	74.3	1.5	94 600	174 000	3 000	**LRBZ 324028 B**
57.150 (2 1/4)	**BRI 365628 UU**	**BRI 365628 U**	980	57.150 (2 1/4)	88.900 (3 1/2)	44.450 (1 3/4)	44.700	69.850 (2 3/4)	65	67	80.7	1.5	98 700	189 000	2 500	**LRBZ 364428 B**
63.500 (2 1/2)	**BRI 406028 UU**	**BRI 406028 U**	1 065	63.500 (2 1/2)	95.250 (3 3/4)	44.450 (1 3/4)	44.700	76.200 (3)	71	73	87.0	1.5	105 000	211 000	2 500	**LRBZ 404828 B**
69.850 (2 3/4)	**BRI 446828 UU**	**BRI 446828 U**	1 421	69.850 (2 3/4)	107.950 (4 1/4)	44.450 (1 3/4)	44.700	82.550 (3 1/4)	77	79	99.7	1.5	109 000	227 000	2 500	**LRBZ 445228 B**
76.200 (3)	**BRI 487232 UU**	**BRI 487232 U**	1 738	76.200 (3)	114.300 (4 1/2)	50.800 (2)	51.050	88.900 (3 1/2)	83.5	86	106.1	1.5	142 000	265 000	2 000	**LRBZ 485632 B**
82.550 (3 1/4)	**BRI 527632 UU**	**BRI 527632 U**	1 851	82.550 (3 1/4)	120.650 (4 3/4)	50.800 (2)	51.050	95.250 (3 3/4)	91	93	111.4	1.5	148 000	287 000	2 000	**LRBZ 526032 B**

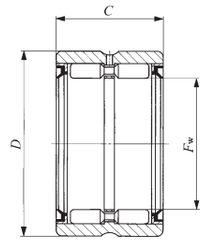
Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The inner ring and the outer ring each have an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

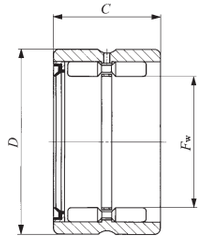
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, Without Inner Ring, Inch Series

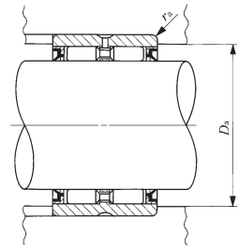




GBR···UU



GBR···U



Shaft dia. 15.875 — 50.800mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			Standard mounting dimensions mm		Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)
	With two seals	With one seal	g	F_w	D	C	D_a Max.	$r_{as\,max}$ (1)	C N	C_0 N	rpm
15.875 (5/8)	GBR 101816 UU	GBR 101816 U	69.5	15.875 (5/8)	28.575 (1 1/8)	25.400 (1)	24.5	0.6	23 500	28 500	5 000
19.050 (3/4)	GBR 122016 UU	GBR 122016 U	79	19.050 (3/4)	31.750 (1 1/4)	25.400 (1)	27	0.6	26 400	34 500	4 000
22.225 (7/8)	GBR 142216 UU	GBR 142216 U	89.5	22.225 (7/8)	34.925 (1 3/8)	25.400 (1)	30	0.6	28 600	40 100	3 500
25.400 (1)	GBR 162416 UU	GBR 162416 U	99	25.400 (1)	38.100 (1 1/2)	25.400 (1)	33.3	0.6	31 000	46 100	3 000
28.575 (1 1/8)	GBR 182620 UU	GBR 182620 U	139	28.575 (1 1/8)	41.275 (1 5/8)	31.750 (1 1/4)	36.3	0.6	43 900	75 300	3 000
31.750 (1 1/4)	GBR 202820 UU	GBR 202820 U	152	31.750 (1 1/4)	44.450 (1 3/4)	31.750 (1 1/4)	39.6	0.6	46 600	83 900	2 500
34.925 (1 3/8)	GBR 223020 UU	GBR 223020 U	163	34.925 (1 3/8)	47.625 (1 7/8)	31.750 (1 1/4)	42.8	0.6	49 500	91 800	2 500
38.100 (1 1/2)	GBR 243320 UU	GBR 243320 U	200	38.100 (1 1/2)	52.388 (2 1/16)	31.750 (1 1/4)	47.3	0.6	54 200	97 700	2 000
41.275 (1 5/8)	GBR 263520 UU	GBR 263520 U	215	41.275 (1 5/8)	55.562 (2 3/16)	31.750 (1 1/4)	50.5	0.6	56 600	105 000	1 900
44.450 (1 3/4)	GBR 283720 UU	GBR 283720 U	230	44.450 (1 3/4)	58.738 (2 5/16)	31.750 (1 1/4)	53.7	0.6	58 900	114 000	1 800
47.625 (1 7/8)	GBR 303920 UU	GBR 303920 U	240	47.625 (1 7/8)	61.912 (2 7/16)	31.750 (1 1/4)	56.2	1	61 100	121 000	1 700
50.800 (2)	GBR 324120 UU	GBR 324120 U	255	50.800 (2)	65.088 (2 9/16)	31.750 (1 1/4)	59.2	1	63 100	130 000	1 600

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.

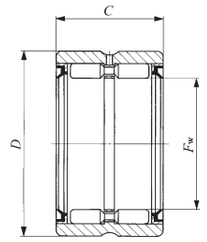
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

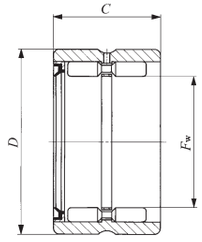
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, Without Inner Ring, Inch Series

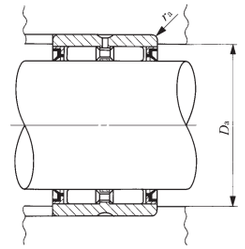




GBR···UU



GBR···U



Shaft dia. 57.150 — 107.950mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			Standard mounting dimensions mm		Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)
	With two seals	With one seal	g	F_w	D	C	D_a Max.	$r_{as\,max}$ (1)	C N	C_0 N	rpm
57.150 (2 1/4)	GBR 364828 UU	GBR 364828 U	515	57.150 (2 1/4)	76.200 (3)	44.450 (1 3/4)	69.2	1.5	87 500	161 000	1 400
63.500 (2 1/2)	GBR 405228 UU	GBR 405228 U	560	63.500 (2 1/2)	82.550 (3 1/4)	44.450 (1 3/4)	75.7	1.5	93 300	179 000	1 300
69.850 (2 3/4)	GBR 445628 UU	GBR 445628 U	610	69.850 (2 3/4)	88.900 (3 1/2)	44.450 (1 3/4)	82	1.5	97 200	197 000	1 100
76.200 (3)	GBR 486028 UU	GBR 486028 U	660	76.200 (3)	95.250 (3 3/4)	44.450 (1 3/4)	88	1.5	101 000	215 000	1 100
82.550 (3 1/4)	GBR 526828 UU	GBR 526828 U	960	82.550 (3 1/4)	107.950 (4 1/4)	44.450 (1 3/4)	99.9	1.5	127 000	231 000	950
88.900 (3 1/2)	GBR 567232 UU	GBR 567232 U	1 240	88.900 (3 1/2)	114.300 (4 1/2)	50.800 (2)	106.3	1.5	170 000	347 000	900
95.250 (3 3/4)	GBR 607632 UU	GBR 607632 U	1 320	95.250 (3 3/4)	120.650 (4 3/4)	50.800 (2)	112.6	1.5	175 000	371 000	850
101.600 (4)	GBR 648032 UU	GBR 648032 U	1 380	101.600 (4)	127.000 (5)	50.800 (2)	119	1.5	182 000	395 000	800
107.950 (4 1/4)	GBR 688432 UU	GBR 688432 U	1 460	107.950 (4 1/4)	133.350 (5 1/4)	50.800 (2)	125.3	1.5	186 000	419 000	750

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.

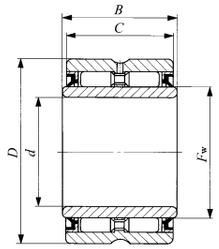
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

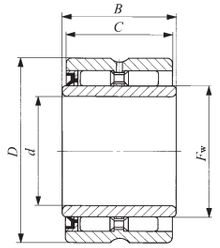
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, With Inner Ring, Inch Series

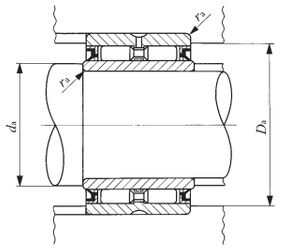




GBRI···UU



GBRI···U



Shaft dia. 9.525 — 44.450mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			B	F_w	Standard mounting dimensions mm				Basic dynamic load rating C	Basic static load rating C_0	Allowable rotational speed(2)	Assembled inner ring
	With two seals	With one seal	g	d	D	C			d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$ (1)	N	N	rpm	
9.525 (3/8)	**GBRI 61816 UU**	**GBRI 61816 U**	94.5	9.525 (3/8)	28.575 (1 1/8)	25.400 (1)	25.650	15.875 (5/8)	14	14.5	24.5	0.6	23 500	28 500	5 000	**LRBZ 61016**
12.700 (1/2)	**GBRI 82016 UU**	**GBRI 82016 U**	110	12.700 (1/2)	31.750 (1 1/4)	25.400 (1)	25.650	19.050 (3/4)	17.5	18	27	0.6	26 400	34 500	4 000	**LRBZ 81216**
15.875 (5/8)	**GBRI 102216 UU**	**GBRI 102216 U**	127	15.875 (5/8)	34.925 (1 3/8)	25.400 (1)	25.650	22.225 (7/8)	21	21.2	30	0.6	28 600	40 100	3 500	**LRBZ 101416**
19.050 (3/4)	**GBRI 122416 UU**	**GBRI 122416 U**	143	19.050 (3/4)	38.100 (1 1/2)	25.400 (1)	25.650	25.400 (1)	24	24.4	33.3	0.6	31 000	46 100	3 000	**LRBZ 121616**
22.225 (7/8)	**GBRI 142620 UU**	**GBRI 142620 U**	200	22.225 (7/8)	41.275 (1 5/8)	31.750 (1 1/4)	32.000	28.575 (1 1/8)	27	27.5	36.3	0.6	43 900	75 300	3 000	**LRBZ 141820**
25.400 (1)	**GBRI 162820 UU**	**GBRI 162820 U**	220	25.400 (1)	44.450 (1 3/4)	31.750 (1 1/4)	32.000	31.750 (1 1/4)	30.5	30.7	39.6	0.6	46 600	83 900	2 500	**LRBZ 162020**
28.575 (1 1/8)	**GBRI 183020 UU**	**GBRI 183020 U**	240	28.575 (1 1/8)	47.625 (1 7/8)	31.750 (1 1/4)	32.000	34.925 (1 3/8)	33.5	33.9	42.8	0.6	49 500	91 800	2 500	**LRBZ 182220**
31.750 (1 1/4)	**GBRI 203320 UU**	**GBRI 203320 U**	286	31.750 (1 1/4)	52.388 (2 1/16)	31.750 (1 1/4)	32.000	38.100 (1 1/2)	37	37.1	47.3	0.6	54 200	97 700	2 000	**LRBZ 202420**
34.925 (1 3/8)	**GBRI 223520 UU**	**GBRI 223520 U**	311	34.925 (1 3/8)	55.562 (2 3/16)	31.750 (1 1/4)	32.000	41.275 (1 5/8)	40.2	40.2	50.5	0.6	56 600	105 000	1 900	**LRBZ 222620**
38.100 (1 1/2)	**GBRI 243720 UU** **GBRI 243920 UU**	**GBRI 243720 U** **GBRI 243920 U**	330 400	38.100 (1 1/2) 38.100 (1 1/2)	58.738 (2 5/16) 61.912 (2 7/16)	31.750 (1 1/4) 31.750 (1 1/4)	32.000 32.000	44.450 (1 3/4) 47.625 (1 7/8)	43.3 43.3	43.4 45	53.7 56.2	0.6 1	58 900 61 100	114 000 121 000	1 800 1 700	**LRBZ 242820** **LRBZ 243020**
41.275 (1 5/8)	**GBRI 264120 UU**	**GBRI 264120 U**	425	41.275 (1 5/8)	65.088 (2 9/16)	31.750 (1 1/4)	32.000	50.800 (2)	48	49	59.2	1	63 100	130 000	1 600	**LRBZ 263220**
44.450 (1 3/4)	**GBRI 284828 UU**	**GBRI 284828 U**	860	44.450 (1 3/4)	76.200 (3)	44.450 (1 3/4)	44.700	57.150 (2 1/4)	52.5	55	69.2	1.5	87 500	161 000	1 400	**LRBZ 283628**

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.

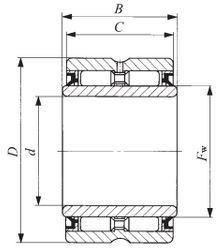
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

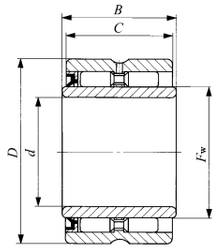
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Seal, With Inner Ring, Inch Series

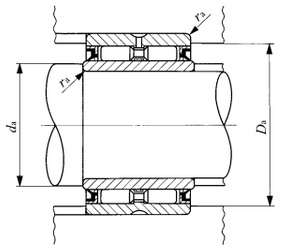




GBRI···UU



GBRI···U



Shaft dia. 50.800 — 95.250mm

Shaft dia. mm (inch)	Identification number		Mass (Ref.)	Boundary dimensions mm(inch)			Standard mounting dimensions mm				Basic dynamic load rating	Basic static load rating	Allowable rotational speed(2)	Assembled inner ring
	With two seals	With one seal	g	d	D	C	B	F_w	d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$(1)	C N	C_0 N	rpm	
50.800 (2)	GBRI 325228 UU	GBRI 325228 U	950	50.800 (2)	82.550 (3 1/4)	44.450 (1 3/4)	44.700	63.500 (2 1/2)	58	61	75.7	1.5	93 300	179 000	1 300	LRBZ 324028
57.150 (2 1/4)	GBRI 365628 UU	GBRI 365628 U	1 050	57.150 (2 1/4)	88.900 (3 1/2)	44.450 (1 3/4)	44.700	69.850 (2 3/4)	65	67	82	1.5	97 200	197 000	1 100	LRBZ 364428
63.500 (2 1/2)	GBRI 406028 UU	GBRI 406028 U	1 140	63.500 (2 1/2)	95.250 (3 3/4)	44.450 (1 3/4)	44.700	76.200 (3)	71	73	88	1.5	101 000	215 000	1 100	LRBZ 404828
69.850 (2 3/4)	GBRI 446828 UU	GBRI 446828 U	1 490	69.850 (2 3/4)	107.950 (4 1/4)	44.450 (1 3/4)	44.700	82.550 (3 1/4)	77	79	99.9	1.5	127 000	231 000	950	LRBZ 445228
76.200 (3)	GBRI 487232 UU	GBRI 487232 U	1 880	76.200 (3)	114.300 (4 1/2)	50.800 (2)	51.050	88.900 (3 1/2)	83.5	86	106.3	1.5	170 000	347 000	900	LRBZ 485632
82.550 (3 1/4)	GBRI 527632 UU	GBRI 527632 U	2 010	82.550 (3 1/4)	120.650 (4 3/4)	50.800 (2)	51.050	95.250 (3 3/4)	91	93	112.6	1.5	175 000	371 000	850	LRBZ 526032
88.900 (3 1/2)	GBRI 568032 UU	GBRI 568032 U	2 130	88.900 (3 1/2)	127.000 (5)	50.800 (2)	51.050	101.600 (4)	97	99	119	1.5	182 000	395 000	800	LRBZ 566432
95.250 (3 3/4)	GBRI 608432 UU	GBRI 608432 U	2 260	95.250 (3 3/4)	133.350 (5 1/4)	50.800 (2)	51.050	107.950 (4 1/4)	103	105	125.3	1.5	186 000	419 000	750	LRBZ 606832

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

●Needle Roller Bearings with Separable Cage - Without Inner Ring

●Needle Roller Bearings with Separable Cage - With Inner Ring

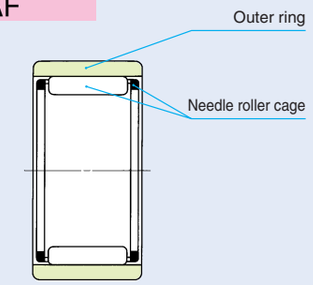


Structure and Features

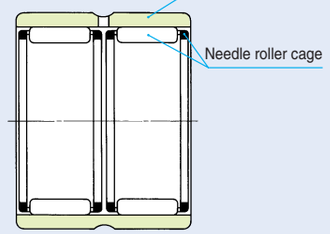
In IKO Needle Roller Bearings with Separable Cage, the inner ring, outer ring and IKO Needle Roller Cage are combined, and they can be separated easily. This type has a simple structure with high accuracy. In addition, the radial clearance can be freely chosen by selecting and combining these component parts. As Needle Roller Cages are used, these bearings have excellent rotational performance.
These bearings are most suitable for mass-production high accuracy products such as machine tools, textile machinery, and printing machines.

Structures of Needle Roller Bearings with Separable Cage

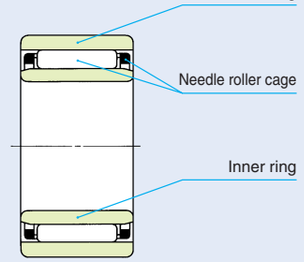
RNAF



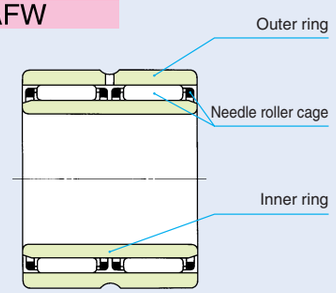
RNAFW



NAF



NAFW



Types

Needle Roller Bearings with Separable Cage are available in the types shown in Table 1.

Table 1 Type of bearing

Type	Single-row		Double-row	
	Without inner ring	With inner ring	Without inner ring	With inner ring
Model code	RNAF	NAF	RNAFW	NAFW

Needle Roller Bearings with Separable Cage - Without Inner Ring

The single-row as well as the double-row types are available with the same sectional height, and either of them can be selected according to load conditions. As shown in the section, "Design of shaft and housing" on page A44, any desired radial internal clearance can be selected by combining a shaft which is heat-treated and finished by grinding.

Needle Roller Bearings with Separable Cage - With Inner Ring

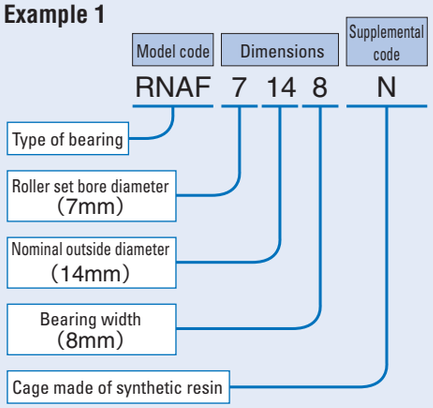
These bearings are made to the CN clearance shown in Table 19 on page A37. When especially high accuracy is required, it is possible to supply semi-finished inner rings which have a finishing allowance on their outside diameter so that they can be ground after being press-fitted to shafts.

Identification Number

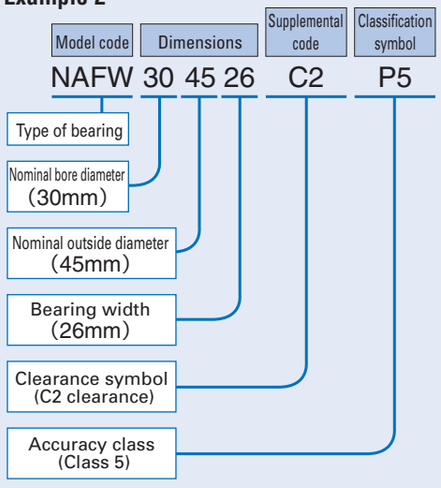
The identification number of Needle Roller Bearings with Separable Cage consists of a model code, dimensions, any supplemental codes and a classification symbol. The arrangement examples are as follows.

Examples of identification number

Example 1



Example 2



Accuracy

Needle Roller Bearings with Separable Cage are manufactured to the accuracy based on JIS (See page A31.). Tolerances for the smallest single roller set bore diameter of bearings without inner ring are based on Table 14 on page A33.

Clearance

Radial internal clearances of Needle Roller Bearings with Separable Cage are made to the CN clearance shown in Table 18 on page A37.

Fit

Recommended fits for Needle Roller Bearings with Separable Cage are shown in Tables 21 to 23 on pages A41 and A42.

Lubrication

Needle Roller Bearings with Separable Cage are not provided with prepacked grease. Perform proper lubrication for use. Using them without lubrication will increase the wear of the rolling contact surfaces and shorten their lives.

Oil Hole

The double-row type outer rings have both an oil hole and an oil groove, but the single-row type outer rings do not. When outer rings with an oil hole are required, attach "-OH" before the clearance symbol in the identification number, and when outer rings with both an oil hole and an oil groove are required, attach "-OG" to the same position.
Example: NAF 203517 - OH C2 P6
When outer rings with multiple oil holes or inner rings with oil hole(s) are required, please contact IKO.

Operating temperature range

For synthetic resin cages, "N" is added at the end of the identification number. The operating temperature range for Needle Roller Bearings with Separable Cage is -20～+120˚C. However, the maximum allowable temperature for synthetic resin cages is +110˚C, and when they are continuously operated, it is +100˚C.

Mounting

Mounting examples of Needle Roller Bearings with Separable Cage are shown in Fig.1.
When mounting Needle Roller Bearings with Separable Cage, it is necessary to locate the needle cage axially. The needle cage is guided by shoulders of the shaft and housing or by side plates, and their guide surfaces must be heat-treated and finished by grinding at right angles to the shaft central axis.
Dimensions related to mounting are shown in the table of dimensions.

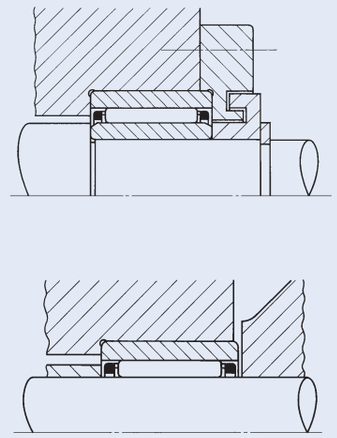


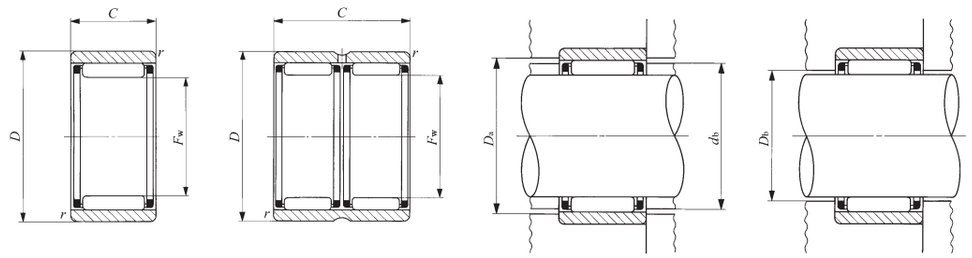
Fig.1 Mounting examples

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

Without Inner Ring







RNAF RNAFW

Shaft dia. 5 — 18mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimensions mm d_b	D_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
5	RNAF 5108N	2.8	5	10	8	0.2	6.7	8.4	5.4	2 420	1 950	85 000
6	RNAF 6138N	5.5	6	13	8	0.3	8.4	11	6.4	2 700	2 320	75 000
7	RNAF 7148N	6.1	7	14	8	0.3	9.4	12	7.4	2 960	2 690	65 000
8	RNAF 81510	8.2	8	15	10	0.3	10.4	13	8.4	3 630	3 600	60 000
	RNAFW 81620	20.5	8	16	20	0.3	10.8	14	8.4	6 220	7 200	60 000
10	RNAF 101710	9.6	10	17	10	0.3	12.4	15	10.4	4 160	4 550	50 000
	RNAF 102012	18.7	10	20	12	0.3	13.5	18	10.4	5 940	6 000	50 000
12	RNAF 122212	19.5	12	22	12	0.3	15.5	20	12.4	9 030	8 460	40 000
14	RNAF 142213	18.7	14	22	13	0.3	17.6	20	14.6	7 860	9 410	35 000
	RNAFW 142220	28.5	14	22	20	0.3	17.6	20	14.6	10 800	14 200	35 000
	RNAF 142612	29	14	26	12	0.3	19.4	24	14.6	9 790	9 680	35 000
15	RNAF 152313	19.7	15	23	13	0.3	18.6	21	15.6	8 250	10 200	35 000
	RNAFW 152320	30.5	15	23	20	0.3	18.6	21	15.6	11 400	15 400	35 000
16	RNAF 162413	21	16	24	13	0.3	19.6	22	16.6	8 620	11 000	30 000
	RNAFW 162420	32	16	24	20	0.3	19.6	22	16.6	11 900	16 700	30 000
	RNAF 162812	31.5	16	28	12	0.3	21.4	26	16.6	10 500	10 900	30 000
17	RNAF 172513	22	17	25	13	0.3	20.6	23	17.6	8 980	11 800	30 000
	RNAFW 172520	33.5	17	25	20	0.3	20.6	23	17.6	12 400	17 900	30 000
18	RNAF 182613	23	18	26	13	0.3	21.6	24	18.6	9 330	12 700	30 000
	RNAFW 182620	35	18	26	20	0.3	21.6	24	18.6	12 900	19 100	30 000
	RNAF 183012	34.5	18	30	12	0.3	23.4	28	18.6	11 800	13 100	30 000
	RNAFW 183024	69.5	18	30	24	0.3	23.4	28	18.6	20 200	26 200	30 000

Notes(1) Minimum allowable value of chamfer dimension r

(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.

Remarks1. The character "N" at the end of the identification number indicates that a synthetic resin cage is incorporated.

2. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.

3. No grease is prepacked. Perform proper lubrication.

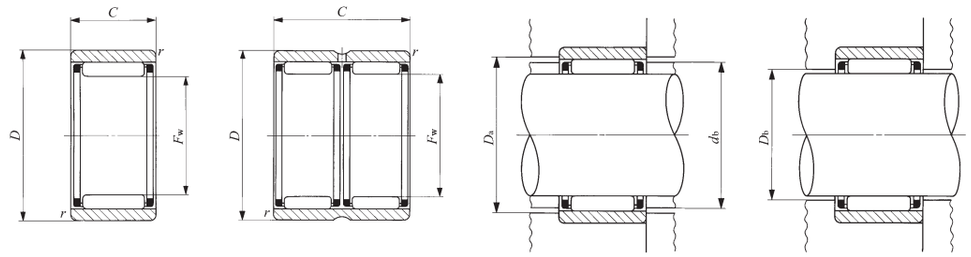
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

Without Inner Ring







RNAF RNAFW

Shaft dia. 20 — 40mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimensions mm d_b	D_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
20	RNAF 202813	25	20	28	13	0.3	23.6	26	20.6	9 590	13 500	25 000
	RNAFW 202826	49.5	20	28	26	0.3	23.6	26	20.6	16 400	27 100	25 000
	RNAF 203212	37.5	20	32	12	0.3	25.4	30	20.6	12 400	14 300	25 000
	RNAFW 203224	75	20	32	24	0.3	25.4	30	20.6	21 200	28 600	25 000
22	RNAF 223013	27	22	30	13	0.3	25.6	28	22.6	10 200	15 200	25 000
	RNAFW 223026	53.5	22	30	26	0.3	25.6	28	22.6	17 500	30 300	25 000
	RNAF 223516	58.5	22	35	16	0.3	27.8	33	22.6	17 600	20 900	25 000
	RNAFW 223532	117	22	35	32	0.3	27.8	33	22.6	30 200	41 800	25 000
25	RNAF 253517	51	25	35	17	0.3	29.5	33	25.6	17 300	26 600	20 000
	RNAFW 253526	78	25	35	26	0.3	29.5	33	25.6	22 400	37 200	20 000
	RNAF 253716	57	25	37	16	0.3	30.4	35	25.6	19 400	24 500	20 000
	RNAFW 253732	114	25	37	32	0.3	30.4	35	25.6	33 200	49 000	20 000
28	RNAF 284016	62.5	28	40	16	0.3	33.4	38	28.6	20 100	26 500	18 000
	RNAFW 284032	125	28	40	32	0.3	33.4	38	28.6	34 400	53 000	18 000
30	RNAF 304017	59	30	40	17	0.3	34.5	38	30.6	18 700	31 100	17 000
	RNAFW 304026	90.5	30	40	26	0.3	34.5	38	30.6	24 200	43 400	17 000
	RNAF 304216	66	30	42	16	0.3	35.4	40	30.6	20 800	28 400	17 000
	RNAFW 304232	132	30	42	32	0.3	35.4	40	30.6	35 700	56 800	17 000
35	RNAF 354517	67.5	35	45	17	0.3	39.5	43	35.6	20 500	36 900	14 000
	RNAFW 354526	103	35	45	26	0.3	39.5	43	35.6	26 600	51 500	14 000
	RNAF 354716	75.5	35	47	16	0.3	40.4	45	35.6	23 100	33 900	14 000
	RNAFW 354732	151	35	47	32	0.3	40.4	45	35.6	39 500	67 800	14 000
40	RNAF 405017	76	40	50	17	0.3	43.5	48	40.8	22 200	42 700	12 000
	RNAFW 405034	152	40	50	34	0.3	43.5	48	40.8	38 000	85 400	12 000
	RNAF 405520	140	40	55	20	0.3	45.2	53	40.8	31 400	48 000	12 000
	RNAFW 405540	280	40	55	40	0.3	45.2	53	40.8	53 900	96 000	12 000

Notes(1) Minimum allowable value of chamfer dimension r

(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.

Remarks1. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.

2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch